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Thoracic limb bones

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Course outline

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- **Scapula**
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- **Forearm skeleton: Ulna, Radius**
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Introduction

The skeleton of the thoracic limb includes the following paired bones: the Humerus (arm skeleton), the Radius and the ulna (forearm skeleton), and the Hand skeleton.
This skeleton is attached to the thorax by the shoulder girdle, formed of two paired bones, the scapula and the clavicle.

Clavicle

It is a long bone, elongated in an extended "S" shape, extending between the Scapula and the Sternum.

Set up :

- At the bottom: the face with a groove running through it;
- Outside: the flattened end;
- Forward: the concave border of this end.

Body :

Present to be described: 2 faces and 2 borders.

1. **Superior surface:**
   It is subcutaneous and palpable.
2. **Inferior surface :**
   Present:
   - In the middle, the groove for the subclavius muscle, which gives insertion to the subclavius muscle;
   - Internally, the costal tuberosity, which provides attachment for the costoclavicular ligament;
   - Outside, the trapezoid line and the conoid tubercle, which give attachment, respectively, to the trapezoid and conoid ligaments.
3. **Anterior border:**
   It is concave in its lateral 1/3, convex in its medial 2/3.
4. **Posterior border:**

It is convex in its lateral 1/3, concave in its medial 2/3.

Extremities:

1. **Acromial end:**
   It is lateral and flattened, it bears the acromial articular surface, and articulates with the acromion of the scapula.
2. **Sternal end:**
   It is medial and rounded, it bears the sternal articular surface, and articulates with the sternum and the first costal cartilage.

Scapula

It is a flat, triangular, dorsal bone, lying against the rib cage. The scapula projects onto the 2nd rib above and the 7th rib below. It articulates with the humerus and the clavicle.

Set up :

- Ahead: the concave face of the bone;
- Outside: the oval articular surface;
- At the top: the notched border.

Faces:

1. **Anterior (costal) surface:**
   It is excavated, forming the subscapular fossa, traversed by 2 or 3 oblique ridges, and gives insertion to the subscapularis muscle.
2. **Posterior (dorsal) surface:**
   It is divided by the scapular spine into two fields, the supraspinous fossa and the infraspinous fossa.
   - Scapular spine: it is a triangular bony plate, perpendicular on the posterior face, and extends outwards into the Acromion, the latter bearing the clavicular articular surface which articulates with the clavicle.
   - Supraspinous fossa: provides insertion for the supraspinatus muscle.
   - Infraspinous fossa: provides insertion for the infraspinatus, teres minor and teres major muscles.

Borders:

- Upper border (cervical): presents the scapular notch, which is surmounted by the transverse ligament of the scapula, and where the suprascapular artery and nerve pass.
- Medial (spinal) border.
- Lateral (axillary) border.

Angles:

- Upper angle.
- Lower angle (tip of the scapula).
- Lateral angle, presents three training programs:
  - *Coracoid process*: it has a shape like a half-flexed finger, it gives insertion to the biceps brachii (short head), coracobrachialis and pectoralis minor muscles.
  - *Glenoid cavity*: the glenoid cavity is an oval-shaped articular surface, notched anteriorly by the glenoid notch, and has the glenoid tubercle in the center. The cavity articulates with the humeral head.
  - *Neck of the scapula*: it is circular and narrowed, it supports the glenoid cavity. The neck has two tubercles, the supraglenoid tubercle, which gives insertion to the biceps brachii muscle (long head), and the infraglenoid tubercle, which gives insertion to the triceps brachii muscle (long head).

Humerus

It is a long bone, which alone forms the skeleton of the arm; it articulates with the scapula at the top and the bones of the forearm at the bottom. It is made up of a diaphysis and two epiphyses.

Set up :

- Up :L'rounded end of the bone;
- Inside:therearticular surface of this extremity;
- Ahead :THEgroove which runs from top to bottom of this extremity.

Diaphysis:

It is cylindrical at the top and triangular and flattened at the bottom. It has three faces and three borders to describe.

1. **Anteromedial surface**: presents the crest of the lesser tubercle which gives insertion to the latissimus dorsi muscles.
2. **Anterolateral view**: presents the deltoid tuberosity (the deltoid "V"), where the deltoid muscle inserts.
3. **Back view**: is crossed by the radial nerve groove, through which the radial nerve passes. It gives insertion to the triceps brachii muscle, the lateral head above the groove, and the medial head below the groove.
4. **Borders**:
   - The anterior one corresponds to the crest of the major tubercle, and is where the major pectoral muscle inserts.
   - The medial
   - The full-back

Proximal epiphysis:

It contains six anatomical structures.

- **Humeral head**: it represents 1/3 of a sphere, with a radius of 30 mm. The head articulates with the glenoid cavity of the scapula.
- **Major tubercle**: is a large lateral protrusion.
- **Minor tuber**: is a small anterior projection.
- **Intertubercular groove**: it separates the two tubercles, major and minor; covered by the transverse humeral ligament, and provides passage for the tendon of the biceps brachii muscle (long head).
- **Anatomical neck**: it separates the humeral head from the two tubercles.
- **Surgical neck**: it separates the proximal epiphysis from the diaphysis, and it is vulnerable to fractures.

Distal epiphysis:

It is flattened, and has three anatomical structures.

1. **Humeral condyle**: it is a joint projection that comprises three articular surfaces:
   - *Chapter*: rounded lateral projection, not visible from the back, surmounted by the radial fossa. It is articulated with the radial head.
   - *Trochlea*: located internally, it is a pulley formed by two cheeks separated by the groove. It is surmounted by two fossae, the coronoid fossa anteriorly and the olecranon fossa posteriorly. It articulates with the trochlear notch of the ulna.
   - *Capitular-trochlear zone* (conoid zone): located between the capitulum and the trochlea.
2. **Medial epicondyle**: it is a very developed and palpable protrusion; it provides attachment for the ulnar collateral ligament of the elbow. It is crossed posteriorly by the groove for the ulnar nerve.

3. **Lateral epicondyle**: it is a poorly developed projection, it provides attachment for the radial collateral ligament of the elbow.

Forearm skeleton

It comprises two long bones, the Ulna and the Radius. These two bones are joined by the proximal radioulnar joint, the distal radioulnar joint and the interosseous forearm membrane.

(1) Ulna

It is the medial bone of the forearm; it articulates with the humeral trochlea above, the radius laterally, and the radioulnar articular disc below. It is formed of a diaphysis and two epiphyses.

Set up :

- Up :L'the largest end;
- Ahead :therehook-shaped articular surface of this end;
- Outside:theresmall articular facet of this extremity.

Diaphysis:

Present to describe three faces and three borders.

- **Faces**: anterior, medial and posterior.
- **Borders**: anterior, posterior and lateral, the latter is called interosseous, where the antebrachial interosseous membrane attaches.

Proximal epiphysis:

It presents two processes, the olecranon and the coronoid process, which together form a "hook" that carries the trochlear notch of the ulna, articular with the humeral trochlea.

1. **Olecranon**: it is a vertical and posterior, subcutaneous process, carries the upper part of the trochlear notch, and gives insertion to the triceps brachii muscle.
2. **Coronoid process**: it is a horizontal and prior process; it presents:
   - Above, the lower part of the trochlear notch;
   - Outside, the radial notch of the ulna, articular with the radial articular circumference;
   - In front, the ulnar tuberosity, where the brachialis muscle inserts.

Distal epiphysis:

It presents, to be described, the ulnar head and the ulnar styloid process.

1. Ulnar head: articulates with the ulnar notch of the radius and the radioulnar articular disc of the wrist.
2. Ulnar styloid process: it is a small posteromedial projection, it gives attachment to the radioulnar articular disc of the wrist and to the ulnar collateral ligament of the carpus.

(2) Radius

It is the lateral bone of the forearm; it articulates with the capitulum above, the ulna medially, and the carpus below. It is formed of a diaphysis and two epiphyses.

Set up :

- Inside: the sharpest border of the bone;
- At the bottom: the largest end;
- Backwards: thereface of this end, which is furrowed.

Diaphysis:

Present to be described, 3 faces and 3 borders.

- **Faces**: anterior, lateral and posterior.
- **Borders**: anterior, posterior and medial, the latter is called interosseous, where the antebrachial interosseous membrane attaches.

Proximal epiphysis:

It is cylindrical and has a radial head, radial neck and radial tuberosity.

- **Radial head**, is made up of:
  - **Radial articular fossa**: superior and concave, articulates with the humeral capitulum and the capitulo-trochlear zone.
  - **Radial joint circumference**: peripheral, articulates with the radial notch of the ulna and the annular ligament.
- **Radial neck**;
- **Radial tuberosity**: ovoid protrusion, where the biceps brachii muscle ends.

Distal epiphysis:

It is a quadrangular pyramid with a bottom base and 4 faces.

- **Side view**: extends downwards by the radial styloid process, the latter being delimited by two grooves through which three muscle tendons pass;
- **Back view**: hollowed out with two grooves, through which seven (7) muscle tendons pass;
- **Medial face**: carries the ulnar notch of the radius which articulates with the ulnar head;
- **Base**: carries the carpal articular surface, which articulates with the radial articular surface of the carpus.

Hand skeleton

It is made up of three parts: the Carpus which corresponds to the wrist, the Metacarpus which forms the skeleton of the palm, and the Phalanges which form the skeleton of the fingers.

Carpus:

It is made up of eight (8) bones, arranged in 2 rows:

- Proximal row: formed from outside in, of the Scaphoid, Lunate, Triquetrum and Pisiform bones;
- Distal row: formed from outside in, of the Trapezium, Trapezoid, Capitate and Hamate bones.

Note: Carpal Tunnel

The anterior face of the carpus is concave, forming the carpal groove, which is completed by the flexor retinaculum forming the carpal tunnel, which provides passage for the median nerve and muscle tendons.

Metacarpus:

It is composed of five (5) metacarpal bones, numbered from "I" to "V", starting from the outside in.

Phalanges:

There are fourteen (14) of them, three phalanges per finger and two phalanges of the thumb. Numbered as follows:

- For the thumb: proximal phalanx (P1) and distal phalanx (P2);
- For the last four fingers: proximal phalanx (P1), intermediate phalanx (P2) and distal phalanx (P3).

Each phalanx is a long bone, having a body, a proximal end, called the base, and a distal end, called the head.

Three points to note:

- **Sesamoid bones**: these are small bones located on the palmar surface of the digital joints.
- **Bone development**: this is done by membranous or cartilaginous ossification. After birth, the bone undergoes two types of growth, in width (at the level of the periosteum) and in length (at the level of the growth plate, also called the conjugate plate).
- **Bone age** (in children and adolescents): It is assessed on an X-ray of the left wrist and hand. It is related to the ossification of the carpal bones and the epiphyses of the long bones of the wrist and hand.